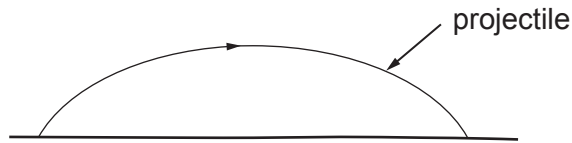


6. (a) A projectile is fired from the Earth's surface and follows a curved path as shown.



Describe and explain how, if at all, the vertical and the horizontal components of velocity of the projectile change during the flight. Ignore the effects of air resistance. [3]

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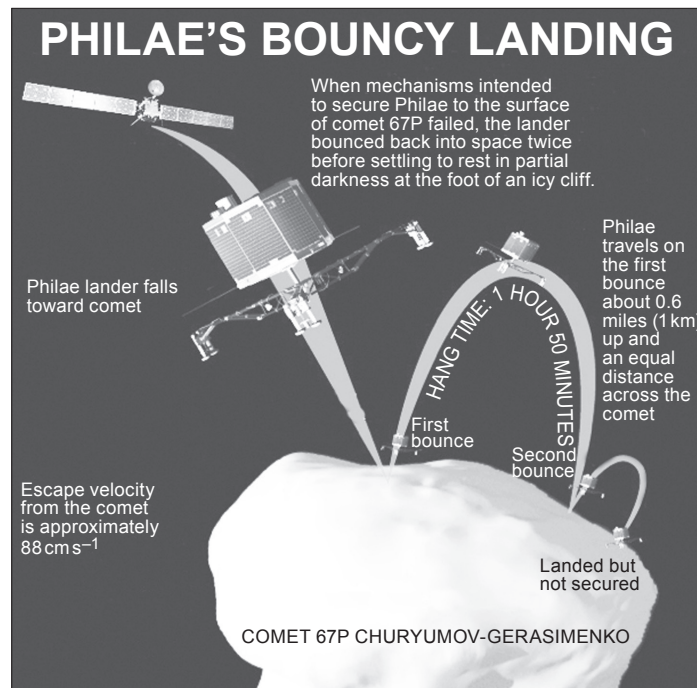
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- (b) In November 2014 the space probe 'Philae' was dropped onto comet 67P. Philae bounced twice before coming to rest. A national newspaper used the following image to describe the landing.



- (i) Use information from the image to calculate the horizontal velocity of the lander between the first and second bounce. [1]

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- (ii) By considering the first bounce, show that the value of the acceleration due to gravity on the comet is approximately 0.0002 m s^{-2} . [2]

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- (iii) Show that the vertical velocity of the lander immediately after the first bounce was greater than 60% of the escape velocity. [3]

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- (c) By considering the following facts discuss whether or not the mission was justified. [3]

- Cost of developing and sending the spacecraft to the comet: £1 billion over 10 years.
- Around 2 000 people involved in the development of the spacecraft and its instruments.
- Advanced solar cell technology developed.
- 28 000 landing announcement 're-tweets' in the first hour.
- Organic molecules detected on comet surface.

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